

Sujal Dave, PhD

POSTDOCTORAL RESEARCHER · DEPARTMENT OF PEDIATRICS (CARDIOLOGY)

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Academic Experience

- Postdoctoral Researcher**, Stanford University
- Oct 2025 - Present Accelerating the open-source package **SimVascular**'s parallel multi-physics finite element solver (*svMultiPhysics*) for computational modeling of the cardiovascular system by linking external libraries and enabling GPU computations.

Education

University of Calgary

Canada

PHD MECHANICAL ENGINEERING (GPA:4.0/4.0)

May 2021 - Jul 2025

- Advisor: Dr. Artem Korobenko
- Advancing the in-house wind turbine simulation tool for faster and accurate wind farm simulations using ALM for environmental flows. Investigating the role of atmospheric stratification on wind turbine wake in different layouts.
- Developed and integrated projection-based reduced-order modeling utility for in-house fluid flow solver written using **FORTRAN** and **OpenMPI** for turbulent flow simulations. Developing neural network based efficiency enhancement for reduced order models using **PyTorch** and **Tensorflow**. Exploring non-intrusive techniques like DMD and PINNs.
- Consistent use of **HPC** clusters available locally (ARC UCalgary) and nationally (Digital Research Alliance of Canada) to run simulations ranging from few thousand elements to couple million elements. Employed remote visualization techniques.
- Utilized a range of specialized software tools, including **ANSA** and **Rhino** for complex meshing, **Visit** and **ParaView** for CFD post-processing and **MATLAB** and **Python** for data analysis.

Scuola Internazionale Superiore di Studi Avanzati (SISSA)

Italy

RESEARCH EXCHANGE STUDENT

Mar 2023 - Aug 2023

- Advisor: Dr. Gianluigi Rozza
- Conducted collaborative research with leading researchers through the Mitacs Globalink research award. Acquired in-depth knowledge of reduced order modeling methodologies and implemented a reduced order modeling framework.
- Explored interdisciplinary research topics on applied mathematics with peers in the field of uncertainty quantification (UQ), scientific machine-learning (SciML) and fractional calculus among others.
- Attended seminars and courses from established professors and learnt basic concepts of new and upcoming simulation tools like **RBniCS** and **PINA**.

KTH Royal University

Sweden

SUMMER SCHOOL ON PHYSICS-INFORMED NEURAL NETWORKS

Jun 2023

- Teachers: Dr. George Em Karniadakis & Dr. Khemraj Shukla
- Completed a two-week intensive summer school on Physics-Informed Neural Networks (PINNs) at KTH Royal Institute of Technology, earning **5 ECTS credits (or 125 study hours)**. Gained hands-on experience with TensorFlow, PyTorch, and **JAX**, with applications to scientific machine learning and PDEs.
- Collaborated with international researchers, resulting in a conference paper and poster presentation at an AI-focused workshop on differential equations in science.
- Developed advanced skills in implementing PINNs for various applications, including UQ and multi-GPU training, utilizing tools like **DeepXDE** and NVIDIA **Modulus**.

North Carolina State University

United States of America

MS (THESIS) MECHANICAL ENGINEERING (GPA:3.9/4.0)

Aug 2018 - Dec 2020

- Advisor: Dr. Clement Kleinstreuer
- Developed a hybrid Computational Fluid-Particle Dynamics (CFPD) and Physiologically Based Pharmacokinetic (PBPK) model to predict drug transfer from the nasal cavity to the bloodstream.
- Demonstrated high accuracy of the model with a deviation of less than 7% from in vivo data for various corticosteroids, providing robust predictions for drug absorption and plasma concentrations.
- Explored the impact of drug solubility and partition coefficients on plasma concentration, showcasing the model's ability to inform nasal drug formulation and dosage optimization.
- Conducted simulations by successfully integrating fluid-particle dynamics with pharmacokinetics for patient-specific intranasal drug delivery analysis.
- Utilized specialized software tools, including **OpenFOAM** as a CFD solver and **ANSYS-ICEM** to mesh nasal and lung geometries.

Institute of Technology, Nirma University

India

B.TECH MECHANICAL ENGINEERING (GPA:8/10)

Aug 2014 - May 2018

- Minor in Finance
- Undergrad research advisor: Dr. Sanjay Jain
- Advanced the low-speed wind tunnel at the university by installing new smoke visualization techniques.
- Conducted computational based research as undergrad final project on the topic "Design and efficiency of centrifugal compressor" and presented at the university-level conference organized by Taylor & Francis.
- Utilized specialized software tools **ANSYS-Fluent** and **ANSYS-CFX** for computing flow field of a centrifugal pump.

Professional Experience

Mar-Apr 2021	Multiphysics Developer Intern , BosonQ Psi (BQP) Private Ltd. Conducted foundational research on applying quantum computing techniques to linear solvers for CFD simulations and developed a finite element utility from scratch for linear elasticity applications.
May-Jun 2017	CFD Summer Intern , Ingersoll Rand (India) Ltd. Gained hands-on experience in CFD using ANSYS to mesh and simulate internal flow within a centrifugal compressor and optimized compressor performance by using wet compression technology.
Dec 2016	Mechanical Engineering Intern , Reliance Industries Ltd. Assisted in the maintenance of a planetary gear pump and an induced draft cooling tower, gaining practical skills in the mechanical workshop.

Publications

PUBLISHED

- Dave S**, Korobenko A. 2025. **Consistent reduced order modeling for wind turbine wakes using variational multiscale method and actuator line model**. *Computer Methods in Applied Mechanics and Engineering*.
- Dave S**, Korobenko A. 2025. **Predicting smooth body flow separation with finite-element-based variational multiscale formulation**. *Computational Mechanics*.
- Jnini A, Goordoyal H, **Dave S**, Korobenko A, Vella F, Fraser K. 2024. **Physics-constrained DeepONet for surrogate CFD models: a curved backward-facing step case**. *ICLR 2024 workshop on AI4DifferentialEquations in Science*.
- Dave S**, Korobenko A. 2023. **Modelling smooth-body flow separation with variational multiscale method, finite elements and weakly enforced Dirichlet boundary conditions**. *14th International ERCOFTAC symposium on engineering turbulence modeling and measurements*.
- Larsson J, Bermejo-Moreno I, Garmann D, Rizzetta D, Baurle R, Mukha T, Toosi S, Schlatter P, Brehm C, Ganju S, Kahraman AB, van Noordt W, Wang ZJ, Duan Z, Blind M, Beck A, **Dave S**, Korobenko A, Strasser W, Zangeneh R, Guzik S, Walters S, Galbraith MC. 2023. **Summary of the smooth body separation test case at the 2022 high fidelity CFD verification workshop**. *2022 AIAA SciTech Forum and Exposition*.
- Dave S**, Kleinstreuer C, Chari S. 2022. **An effective PBPK model predicting dissolved drug transfer from a representative nasal cavity to the blood stream**. *Journal of Aerosol Science*.

IN PREP

Zhao H, Kaiser AD, Kong F, Brown AL, Hu Z, Codoni D, **Dave S**, Marsden AL. [Comparative analysis of resistive immersed surface and immersed boundary methods for aortic valve simulation](#).

Codoni D, **Dave S**, Parker DW, Brown AL, Vallejos JJ, Zhao H, Adil D, Sexton ZA, Pfaller MR, Bucelli M, Shadden SC, Taylor CA, Marsden AL. [svMultiPhysics: a finite element-based solver for cardiovascular simulations](#).

Dave S, Regmi A, Stabile G, Rozza G, Korobenko A. [Parametric reduced order modeling of wind turbine flows in the variational multiscale framework](#).

Regmi A, **Dave S**, Korobenko A. [ALM-VMS-ROM for wind turbines in stratified atmospheric flows](#)

Scholarships & Awards

2025	Departmental Conference Award , MME Department, University of Calgary	CA\$ 750
2024	Alka and Sanjeev Khanna Graduate Scholarship , MME Department, University of Calgary	CA\$ 10,000
	Departmental Conference Award , MME Department, University of Calgary	CA\$ 500
	Travel Award , Canadian Association for Computational Science and Engineering (CACSE)	CA\$ 750
2023	Eyes High International Doctoral Scholarship , University of Calgary	CA\$ 15,000
	Mitacs Globalink Research Award , Mitacs Canada	CA\$ 6,000
2021	Alka and Sanjeev Khanna Graduate Scholarship , MME Department, University of Calgary	CA\$ 10,000

Conference Presentations

Dave S, Korobenko A. 2025. [ROM-VMS formulation for environmental flows and wind energy applications](#). 3rd IACM Digital Twins in Engineering Conference (DTE 2025), Paris, France.

Dave S, Korobenko A. 2024. [Reduced order modeling of wind turbines using variational multiscale method](#). Canadian Wind Energy Research Network (CWERN) Fall 2024 Meeting, Calgary, Canada.

Dave S, Korobenko A. 2024. [Projection-based model order reduction of environmental flows using FEM-based variational multiscale method](#). 16th World Congress on Computational Mechanics (WCCM), Vancouver, Canada.

Dave S, Korobenko A. 2024. [Projection-based model order reduction of turbulent flows](#). Mechanical and Manufacturing Graduate Student Conference, Calgary, Canada.

Dave S, Korobenko A. 2024. [Physics-constrained DeepONet for surrogate CFD models: a curved backward-facing step case](#). Poster Presentation at ICLR 2024 Workshop on AI4DifferentialEquations in Science, Vienna, Austria.

Dave S, Korobenko A. 2023. [Modeling smooth-body flow separation with variational multiscale method, finite elements and weakly enforced Dirichlet boundary conditions](#). 14th International ERCOFTAC symposium on engineering turbulence modeling and measurements, Barcelona, Spain.

Dave S, Korobenko A. 2023. [Variational multiscale framework with Isogeometric analysis for smooth body flow separation in aerospace and environmental applications](#). 1st International Conference on Math 2 Product (M2P), Taormina, Italy.

Dave S, Korobenko A. 2023. [Reduced order modeling with variational multiscale method](#). 22nd Computational Fluids Conference (CFC), Cannes, France.

Dave S, Korobenko A. 2022. [Predicting smooth body flow separation using isogeometric analysis and variational multi-scale method](#). 10th International Conference on Isogeometric Analysis (IGA), Banff, Canada.

Dave S, Korobenko A. 2022. [Application of variational multiscale methods and weakly enforced Dirichlet boundary condition for smooth-body separation](#). 1st High-Fidelity CFD Workshop at AIAA SciTech 2022, San Diego, USA.

Dave S, Korobenko A. 2022. [Predicting smooth body flow separation using variational multiscale method and weak-Dirichlet boundary conditions](#). Mechanical and Manufacturing Graduate Student Conference, Calgary, Canada.

Teaching Experience

Winter 2025	Computational Fluid Dynamics , Teaching Assistant (<i>taught ANSA & OpenFOAM</i>)
Fall 2024	Aerodynamics , Teaching Assistant (<i>taught ANSYS-Fluent</i>)
Winter 2024	Computational Fluid Dynamics , Teaching Assistant (<i>taught ANSA & OpenFOAM</i>)
Fall 2023	Finite Element Method , Teaching Assistant (<i>taught ANSA</i>)
Fall 2022	Aerodynamics , Teaching Assistant (<i>taught ANSYS-Fluent</i>)
Winter 2022	Fundamental of Fluid Mechanics , Teaching Assistant
Spring 2020	Foundations of Fluid Dynamics , Grading Assistant

Mentoring

2025	Ciara Browne , Undergraduate Student, University of Toronto
2024-2025	Manal Chaudhry , Undergraduate Student, University of Calgary
2024-2025	Olivia Ikeri , Undergraduate Student, University of Calgary
2024-2025	April Bushrod , Undergraduate Student, University of Calgary
2024	Hannah Blakeley , Undergraduate Student, University of Calgary

Outreach & Professional Development

SERVICE AND OUTREACH

2023-2024	Women in Science and Engineering (WISE) Research Night , Research Ambassador
2016-2017	Mechanical Engineering Student Association (MESA) , General Secretary
2015	Make A Difference (MAD) NGO , Volunteer